

- 9 A printer runs out of ink and needs a new ink cartridge. An interrupt is sent and a message is generated to notify the user that the ink cartridge is empty.

(a) Complete the paragraph about the use of an interrupt in this process.

Use the terms from the list.

Some of the terms in the list will **not** be used. You should only use a term once.

binary	clock	computer	data	error
fetch–decode–execute cycle	hexadecimal	higher	interrupt queue	
interrupt service routine (ISR)	lower	printer	priority level	
secondary storage	sorting	transmission		

The sends an interrupt to the
 The interrupt is given a
 The processor completes its current
 and checks the
 to see if there is an interrupt with
 priority than the current task. If there is, it stores
 the current task and fetches the interrupt. It then calls the
 , which is a sequence of instructions that
 handles the interrupt. The interrupt is handled, which generates a message to the user to
 notify them that the ink cartridge is empty.

[7]

(b) Handling interrupts is one function of an operating system.

Give **two** functions of an operating system that relate to memory.

1

2

[2]